

**Language Doesn't Matter: Evaluating the Efficacy of Large Language Models in Analyzing  
Sentiment in French Textual Data**

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### Abstract

This study examines whether general-purpose large language models (LLMs) can analyze sentiment in French text without task-specific fine-tuning, and whether prompt language affects their performance. Twelve open- and closed-weight LLMs are evaluated on 200 French sentences drawn from a corpus of 2,683 French news articles on open-source software, and compared with dictionary-based methods across three conditions: French prompts on French text, English prompts on French text, and English prompts on machine-translated English text. Prompt language has no statistically meaningful effect on performance, and equivalence tests rule out any substantial advantage of English prompts, contrary to the English-prompting advantage reported in prior work; machine-translating the text into English, by contrast, lowers performance. Leading closed-weight models achieve strong performance ( $r > 0.80$ ), larger open-weight models offer a viable, lower-cost alternative, and the smallest model tested (1B parameters) fails to follow the task instructions. LLMs outperform a standard sentiment dictionary on correlation with human judgments and on coarse classification, though the dictionary offers greater transparency. All models perform substantially better on three-category (positive/neutral/negative) than on seven-category classification. These findings suggest social science researchers can use general-purpose LLMs for sentiment analysis of French text with either French or English prompts, without sacrificing accuracy, but should analyze the original text rather than an English translation.

*Keywords:* Sentiment Analysis, Large Language models, Cross-lingual NLP, French Language, Open-Source models, Computational Social Science, Prompt Engineering

## Language Doesn't Matter: Evaluating the Efficacy of Large Language Models in Analyzing Sentiment in French Textual Data

### Introduction

Most foundational large language models (LLMs) are trained predominantly on English data (Touvron et al., 2023), and a growing body of research reports that prompting in English improves performance on non-English tasks, even when the target text itself is non-English (Jiang et al., 2020; Lin et al., 2022; Muennighoff et al., 2023; Nie et al., 2024). If this “English prompt advantage” holds generally, it imposes a real cost on non-anglophone researchers, who would need to translate their prompts to get the best results from otherwise capable models.

This paper puts that premise to the test in the case of French. Sentiment analysis of French-language text offers a demanding case: sentiment carries cultural and linguistic nuance that machine translation can alter (Mohammad et al., 2016); it is a common measurement task in political communication and the social sciences more broadly (van Atteveldt et al., 2021; Young & Soroka, 2012); it admits both continuous (intensity) and categorical (polarity) evaluation (Young & Soroka, 2012); and established dictionary-based methods, including a French adaptation of the Lexicoder Sentiment Dictionary, offer a meaningful performance benchmark (Duval & Pétry, 2016; van Atteveldt et al., 2021; Young & Soroka, 2012). Although limited to a single language, the case bears on the broader premise that prompting in English is the safer choice whatever the language of the text.

Testing this premise presupposes that general-purpose LLMs — used off the shelf, without task-specific fine-tuning — can analyze sentiment in a French-language corpus competently in the first place, whether prompted in French or in English. Specialized models such as BERT variants achieve strong performance through fine-tuning (Anirban & Abhik, 2021; Michailidis, 2024; Rafeef A. Hameed et al., 2023), but require machine learning expertise, computational resources, and labeled data that many social science researchers lack (Chalkidis et al., 2020; Laurer et al., 2023; Strubell et al., 2019). General-purpose LLMs accessed through APIs offer a more accessible alternative, requiring primarily prompting skills (Ghosh et al., 2023; Perron et al.,

2024) — but only if they perform adequately without fine-tuning in the first place.

This study therefore asks two questions: (1) does prompt language (French vs. English) meaningfully affect sentiment analysis performance on French text, and (2) can general-purpose LLMs — open and closed-weight<sup>1</sup> — perform this task well enough, without fine-tuning, to be a viable alternative to dictionary-based methods for social science research? For the first question, the literature above implies a directional expectation, derived from prior work rather than preregistered: *H1: English prompts outperform French prompts on French text*. The second question is descriptive and carries no hypothesis. The results do not support H1, and several models — including open-weight ones — outperform a traditional dictionary method.

### Literature Review

Specialized, fine-tuned models achieve strong multilingual sentiment analysis performance — often above 85-95% accuracy across Indo-European, Semitic, and Asian languages (Ahmadian et al., 2024; Anirban & Abhik, 2021; ElJundi et al., 2019; Michailidis, 2024; Rafeef A. Hameed et al., 2023), including for lower-resource languages (Nazir et al., 2025; Salahudeen et al., 2023; Tela et al., 2024) and via cross-lingual transfer (Kumar & Albuquerque, 2021; Pan et al., 2023). This performance is costly: such models require large parameter counts (Francesco et al., 2021; Rafeef A. Hameed et al., 2023), fine-tuning expertise (Kotelnikova et al., 2023; Přibáň et al., 2024), and substantial labeled training data (Mikhail et al., 2024; Salahudeen et al., 2023) — barriers that put them out of reach for many social science researchers.

By contrast, general-purpose foundational LLMs used without fine-tuning remain understudied for multilingual sentiment analysis. Preliminary work with Mistral-7B, LLaMA2, and instruction-tuned models suggests promise in Italian, Russian, and cross-lingual settings (Buscemi & Proverbio, 2024; Plagwitz et al., 2024; Rusnachenko et al., 2024; Venerito & Iannone, 2024). Evidence on prompt language effects is mixed: some studies find language-specific

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<sup>1</sup> While “open source” traditionally implies access aligned with FOSS principles, its usage within the LLM context often refers more specifically to the public release of model weights, a condition more accurately described as “open weights.” This paper adopts this common LLM terminology while acknowledging that most models in the “open source” category are technically “open weights.”

prompting advantages (Hämäläinen et al., 2022; Vileikytė et al., 2024), while others report that English prompting broadly improves cross-linguistic performance (Jiang et al., 2020; Lin et al., 2022; Muennighoff et al., 2023; Nie et al., 2024). This study addresses this gap by evaluating leading open- and closed-weight foundational LLMs on French text.

## Data & Methods

### Data Collection and Preparation

The corpus consists of 2,683 French-language news articles from 13 media sources (9 from Québec, 1 from Ontario, 3 from France), an exhaustive collection of articles discussing open-source software from the Eureka database’s “Major daily newspapers (FR)” category. Articles were selected using the search query below, over the period from January 1, 1991 — the release of the Linux kernel, the largest open-source project to date — to January 1, 2025. The start date excludes nothing in practice: the earliest matching article was published June 23, 1995.

```
TEXT= ("logiciel libre" | "logiciels libres" | "open source" | "open-source" | "logiciel  
open source" | "logiciels open source" | "code source ouvert" | "software libre"|  
"free software" | "code source libre" | "Free Software Foundation" | "Richard  
Stallman")
```

Table 1 shows the distribution of articles by source media.

Articles were downloaded in HTML format with a Python/Selenium scraper (rather than as Eureka’s default PDF batches, which are slow to retrieve and poorly suited to text analysis), and publication metadata and full-text content were extracted with custom R functions.

**Table 1**  
*Summary of Collected News Articles Discussing Open Source Software by Source Media*

| source_media                         | article_count | earliest_date | latest_date |
|--------------------------------------|---------------|---------------|-------------|
| Libération                           | 769           | 1995-06-23    | 2024-11-04  |
| Monde, Le                            | 666           | 2001-01-09    | 2024-12-26  |
| Figaro, Le                           | 402           | 1998-10-21    | 2024-12-04  |
| Devoir, Le                           | 318           | 1998-07-20    | 2024-10-26  |
| Soleil, Le (Québec, QC)              | 179           | 1998-11-08    | 2023-11-18  |
| Presse, La                           | 152           | 1998-03-26    | 2017-10-14  |
| Journal de Montréal, Le              | 65            | 2006-12-20    | 2023-04-24  |
| Journal de Québec, Le                | 45            | 2007-03-24    | 2024-10-19  |
| Tribune, La (Sherbrooke, QC)         | 43            | 2000-03-29    | 2022-12-03  |
| Droit, Le (Ottawa, ON)               | 17            | 1998-11-10    | 2014-07-17  |
| Nouvelliste, Le (Trois-Rivières, QC) | 13            | 2000-01-13    | 2022-03-05  |
| Quotidien, Le (Saguenay, QC)         | 10            | 2008-11-01    | 2019-04-10  |
| Voix de l'Est, La (Granby, QC)       | 4             | 2010-09-17    | 2015-02-14  |

**Figure 1**  
*Temporal Distribution of News Articles Discussing Open Source Software*

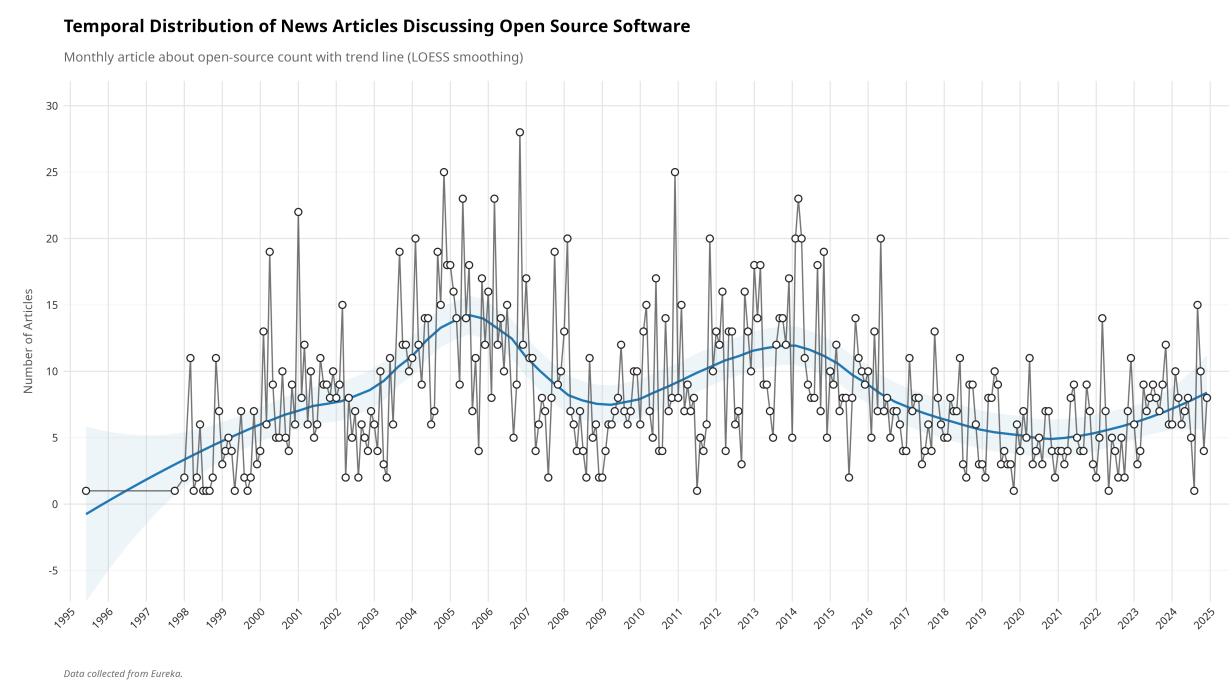


Figure 1 shows the temporal distribution of articles, with notable increases in coverage in the early 2000s corresponding to major open-source developments.

## Translation

To enable the EN→EN condition described below, the French corpus was machine-translated into English using Google Translate's document translation service, processed in batches via an automated R workflow that preserved document IDs to keep French and English versions aligned in the final dataset.<sup>2</sup>

## Manual Annotation

A validation benchmark of 200 sentences was randomly sampled from the corpus, and each sentence was machine-translated into English via Google Translate and the `polyglotR` package. This sample size falls within the 100 to 300 units generally recommended for validating automated sentiment measures against a manually coded gold standard ([van Atteveldt et al., 2021](#)), and yields 95% confidence intervals of roughly  $\pm 0.05$  around a correlation of 0.80. Three independent coders rated each sentence's sentiment on a scale from -1 (very negative) to 1 (very positive) using a custom Shiny application with keyboard shortcuts and scale reminders, and the three ratings were averaged to obtain the ground truth used to evaluate both dictionary-based and LLM methods.

Agreement between individual coders was moderate. Krippendorff's alpha (interval) was 0.597 (95% bootstrap CI [0.516, 0.656]), the mean pairwise correlation between coders was 0.687, and all three coders assigned the same sign (negative, zero, or positive) to only 42.5% of sentences. Two features of the task account for much of this: coders used a fine-grained scale (more than 60 distinct values), and most sentences carry mild sentiment, so small differences often cross zero. Because the ground truth averages three coders, it is considerably more reliable than any single coder: the Spearman-Brown projection from the mean pairwise correlation puts the reliability of the three-coder mean at 0.868.

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<sup>2</sup> See `src/21_translate_to_english.R` in the project repository for the full translation pipeline.

## Dictionary-Based Sentiment Analysis

Lexicon-based sentiment scoring was applied to both the French original and the English translation as a classical baseline. For French text, a customized version of the French Lexicoder Sentiment Dictionary (`fr1sd`) (Duval & Pétry, 2016) was applied to lowercased, stopword-removed, sentence-segmented text, scoring each sentence as  $(\text{positive terms} - \text{negative terms}) / \text{total words}$ . For English translations, the standard Lexicoder Sentiment Dictionary (Young & Soroka, 2012) was used with additional handling for contractions, proper nouns, and negation (e.g., “not good” scored as negative), scoring each sentence as  $((\text{positive} + \text{negated\_negative}) - (\text{negative} + \text{negated\_positive})) / \text{total words}$ .

## LLM-Based Sentiment Analysis

Twelve general-purpose foundational LLMs were evaluated, selected for prominence and affordability — realistic choices for researchers who need to analyze large corpora rather than a single sample. Specialized fine-tuned models can outperform them on narrow tasks, but at the cost in expertise and infrastructure set out above.

The evaluation spans nine open-weight models — Llama 3.2 1B, Llama 3.2 3B, Llama 3.1 8B, GPT-OSS 20B, Qwen3 32B, Llama 4 Scout, GPT-OSS 120B, Qwen3 235B-A22B, and DeepSeek V3.2, ranging from 1B to 671B total parameters — and three closed-weight models: Claude Haiku 4.5 (Anthropic), Gemini 3.5 Flash (Google), and GPT-5.6 Luna (OpenAI). This lineup updates an initial 2024-2025 evaluation of 11 comparable models using an identical experimental design; several models from that original evaluation (e.g., GPT-4o, Gemini 2.0 Flash) had already been deprecated by their providers within eighteen months, underscoring how quickly this landscape moves. The smallest model, Llama 3.2 1B, rarely followed the instruction to answer with a number: about one call in 23 returned a parseable score, and even after retries it produced a usable score for only 58 to 74 of the 200 sentences, depending on the condition. It was retained deliberately: while models of around 8B parameters can handle the task, a 1B-parameter model appears too small to follow the instructions reliably at all, which marks a practical lower bound on model size. Its metrics rest on a reduced and non-random subset of sentences and are

not directly comparable to those of the other models.

Each model was tested under three conditions to isolate the effects of prompt language and text language: French prompts on French text (FR→FR), English prompts on French text (EN→FR), and English prompts on the machine-translated English text (EN→EN).

All prompts followed a standardized structure: a task definition, a five-point sentiment scale from -1.0 to 1.0 with anchor descriptions (allowing intermediate values), an instruction to consider cultural and linguistic nuance, and a constraint to respond with only a numerical value. French prompts were direct translations of the English prompts. The English-prompt/French-text version is shown below.<sup>3</sup>

```
en_prompt_fr_text <- function(text) {  
  paste0("Please analyze the sentiment of the following French text and provide a single  
    numerical rating according to this scale:  
Sentiment Scale:  
-1.0: Strong negative sentiment - highly critical, hostile, or pessimistic content  
-0.5: Moderate negative sentiment - somewhat negative, disapproving, or concerned  
    content  
0.0: Neutral sentiment - factual, balanced, or neither positive nor negative content  
0.5: Moderate positive sentiment - somewhat positive, approving, or optimistic content  
1.0: Strong positive sentiment - highly supportive, enthusiastic, or optimistic content  
You may also use values between these points for intermediate sentiment levels  
(e.g., -0.7, 0.3).  
Important instructions:  
1. First, carefully read and understand the text, considering cultural and linguistic  
    nuances in French.  
2. Analyze the emotional tone, word choice, and overall message.  
3. Respond ONLY with a single numerical value between -1.0 and 1.0 that best  
    represents the sentiment.  
4. Do not include ANY explanations, analysis, or additional text in your response.  
Here is the text to analyze: ", text)  
}
```

Each model was queried via the `ellmer` R package, using the vendors' own APIs for the three closed-weight models and OpenRouter for the nine open-weight models, with responses

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<sup>3</sup> The French-prompt/French-text version is identical except for language; see `src/93_prompts.R` in the project repository for the exact prompts, and `src/40_prompt.R` for the full evaluation pipeline, including model initialization, retry logic, rate limiting, and checkpointing.

capped at 400 tokens. Because OpenRouter can serve the same model through several providers at different numerical precisions, each open-weight model was pinned to a single named provider endpoint, at the highest precision offered, with fallback to other providers disabled.<sup>4</sup> Gemini 3.5 Flash was queried with its default settings, under which it generates reasoning tokens before answering (354 output tokens per call on average, against 7 for Claude Haiku 4.5). Each sentence was scored three times per model-condition combination (21,600 scores in total), and the mean of the three runs was taken as the final score. Despite explicit instructions to respond with a bare number, models frequently returned formatted text (e.g., “Sentiment: 0.5”); these responses were parsed with a pattern-matching function validated against the expected -1.0 to 1.0 range.

### Validation and Performance Evaluation

Performance was assessed with five complementary metrics: Pearson correlation between each model’s continuous predictions and the manually annotated ground truth; Lin’s concordance correlation coefficient (CCC), which, unlike Pearson correlation, also penalizes differences in scale and location, and so captures whether predictions reproduce the intensity of human ratings rather than only their ordering; mean absolute error (MAE); weighted F1 score on a 7-category classification (“very\_negative” through “very\_positive”); and weighted F1 score on a simplified 3-category classification (negative/neutral/positive) that groups the 7 categories into broad polarity bands. Weighted (rather than macro) F1 is used because the manually coded categories are unevenly distributed — “somewhat\_positive” alone accounts for 44% of annotated sentences, while “very\_negative” and “very\_positive” each account for 1% or less — so a weighted average better reflects overall classification quality given this class imbalance. Comparing the two classification schemes separates fine-grained sentiment discrimination (7-category) from broad polarity classification (3-category). All metrics were calculated on complete cases only.

Condition comparisons are paired by model. Alongside paired t-tests, two one-sided tests (TOST) against an equivalence bound of  $\pm 0.05$  on each metric’s scale assess whether a difference

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<sup>4</sup> Endpoint pins, and the reasoning settings used to suppress chain-of-thought in models that reason by default, are listed in `src/94_models_map.R`.

is small enough to be treated as negligible, since a non-significant t-test cannot on its own establish the absence of an effect. These comparisons and the averages reported in the Results cover the eleven models that scored all 200 sentences (33 model-condition combinations) and exclude the dictionary baselines, which have no prompt. Llama 3.2 1B is reported separately: it scored a different, partial subset of sentences in each condition, so its differences across conditions would compare unlike samples.

## Results

### **Prompt Language Has No Meaningful Effect; Machine-Translating the Text Does**

Holding the text constant, prompt language made no detectable difference. French prompts on French text (FR→FR) edged out English prompts on the same text (EN→FR) on every metric — mean correlation 0.700 vs. 0.673, MAE 0.209 vs. 0.217, 3-category F1 0.669 vs. 0.646, 7-category F1 0.364 vs. 0.347 — but none of these paired differences was statistically significant (all  $p > 0.12$ ). Equivalence tests go further: FR→FR and EN→FR were statistically equivalent within  $\pm 0.05$  on MAE ( $p = 0.002$ ), 7-category F1 ( $p = 0.034$ ), and 3-category F1 ( $p = 0.034$ ). Correlation was the one metric where equivalence could not be established: the 90% confidence interval of the difference,  $[-0.032, 0.085]$ , leaves room for a French-prompt advantage of up to 0.085, but not for an English-prompt advantage larger than 0.032. H1 is therefore not supported.

Machine-translating the text into English, by contrast, had a measurable cost. Relative to FR→FR, scoring the English translation (EN→EN) lowered mean correlation by 0.053 (0.700 vs. 0.647,  $p = 0.007$ ) and 3-category F1 by 0.063 (0.669 vs. 0.606,  $p < 0.001$ ); the 3-category loss also held against EN→FR (0.041,  $p = 0.005$ ). Differences in MAE and 7-category F1 across all three conditions remained within the  $\pm 0.05$  equivalence bound. The translation penalty is therefore concentrated in how well models rank sentences by sentiment and identify their polarity, consistent with sentiment altered in machine translation ([Mohammad et al., 2016](#)).

### **LLMs Are a Viable Alternative to Dictionary Methods**

General-purpose LLMs, used without fine-tuning, correlated with human judgments well above the dictionary baselines. The best configurations exceeded  $r = 0.80$ : GPT-5.6 Luna led on

correlation ( $r = 0.810$  in FR→FR) and concordance ( $CCC = 0.779$ ), Claude Haiku 4.5 on both 3-category F1 (0.778) and 7-category F1 (0.537, both FR→FR), and Gemini 3.5 Flash on MAE (0.156 in EN→FR). Nine of the 33 model-condition combinations correlated with the ground truth more strongly than individual human coders agreed with the mean of the other two (mean leave-one-out  $r = 0.748$ , range 0.724–0.779).<sup>5</sup>

**Figure 2**  
*Sentiment Analysis Models Correlation with Ground Truth*

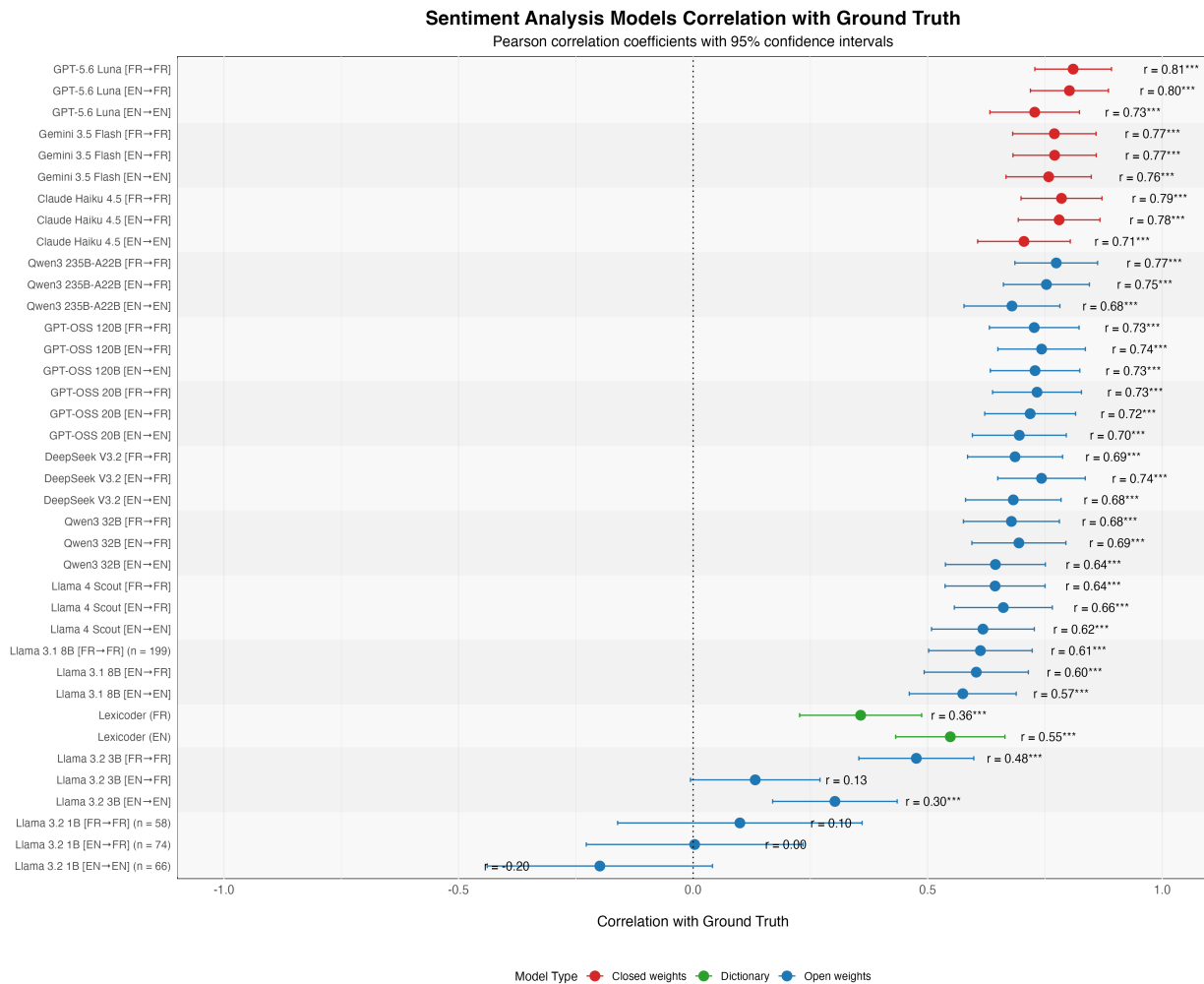


Figure 1. Cross-lingual sentiment analysis performance across model architectures. Models grouped by type (color) and sorted by correlation strength. Prompting mechanisms within clusters ordered as FR→FR (French prompt on French text), EN→FR (English prompt on French text), and EN→EN (English prompt on translated text). Error bars represent 95% confidence intervals; \* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ .

Closed-weight models outperformed open-weight ones on average (mean  $r = 0.768$ )

<sup>5</sup> This comparison favors the models slightly, since the ground truth they are compared against averages three coders rather than two.

vs. 0.638; 3-category F1 0.689 vs. 0.622), but the gap closed at the top of the open-weight range: Qwen3 235B-A22B reached  $r = 0.774$  in FR→FR, above Gemini 3.5 Flash in the same condition, and GPT-OSS 120B and GPT-OSS 20B averaged  $r = 0.733$  and  $0.716$  across conditions. Performance degraded steadily below that. Llama 3.1 8B remained usable ( $r = 0.575$ – $0.612$ ), Llama 3.2 3B was erratic ( $r = 0.476$  in FR→FR but  $0.132$  in EN→FR), and Llama 3.2 1B failed outright: it returned a usable score for only 58 to 74 sentences per condition, and on those its correlation with the ground truth ranged from  $-0.199$  to  $0.100$ .

The English Lexicoder Sentiment Dictionary reached a correlation of  $0.548$  and the French version  $0.357$ , against a mean of  $0.673$  for the eleven LLMs; on 3-category F1 the dictionaries ( $0.588$  and  $0.557$ ) also trailed the LLM mean ( $0.640$ ), though the English dictionary outperformed Llama 3.2 3B and Qwen3 32B. The gap is widest on concordance: the LLMs averaged a CCC of  $0.631$ , against  $0.233$  and  $0.181$  for the English and French dictionaries. On MAE, however, both dictionaries ( $0.203$  and  $0.210$ ) did slightly better than the LLM average ( $0.213$ ). This reflects the dictionaries' compressed output rather than accuracy: their scores cluster near zero (SD =  $0.065$  for the English dictionary, vs.  $0.286$  for the ground truth and  $0.38$ – $0.47$  for the LLMs), which keeps absolute errors small on a sample dominated by mildly positive sentences while discriminating poorly between them.

### **Classification Granularity**

All models performed substantially better on 3-category than 7-category classification (best F1 of  $0.778$  vs.  $0.537$ ; mean  $0.640$  vs.  $0.349$ ; see Figure 3, below). Models most reliably identified positive and negative sentiment (7-category F1 =  $0.482$  and  $0.441$ , rising to  $0.690$  and  $0.627$  in the 3-category scheme) and struggled most with the extreme categories (very\_negative F1 =  $0.256$ , very\_positive F1 =  $0.124$ ), which are nearly absent from the sample (two sentences and one sentence respectively). Neutral remained the hardest 3-category label (F1 =  $0.258$ ). Researchers needing fine-grained sentiment distinctions may find general-purpose LLMs insufficient without further adaptation; the broader positive/neutral/negative distinction is handled far more reliably.

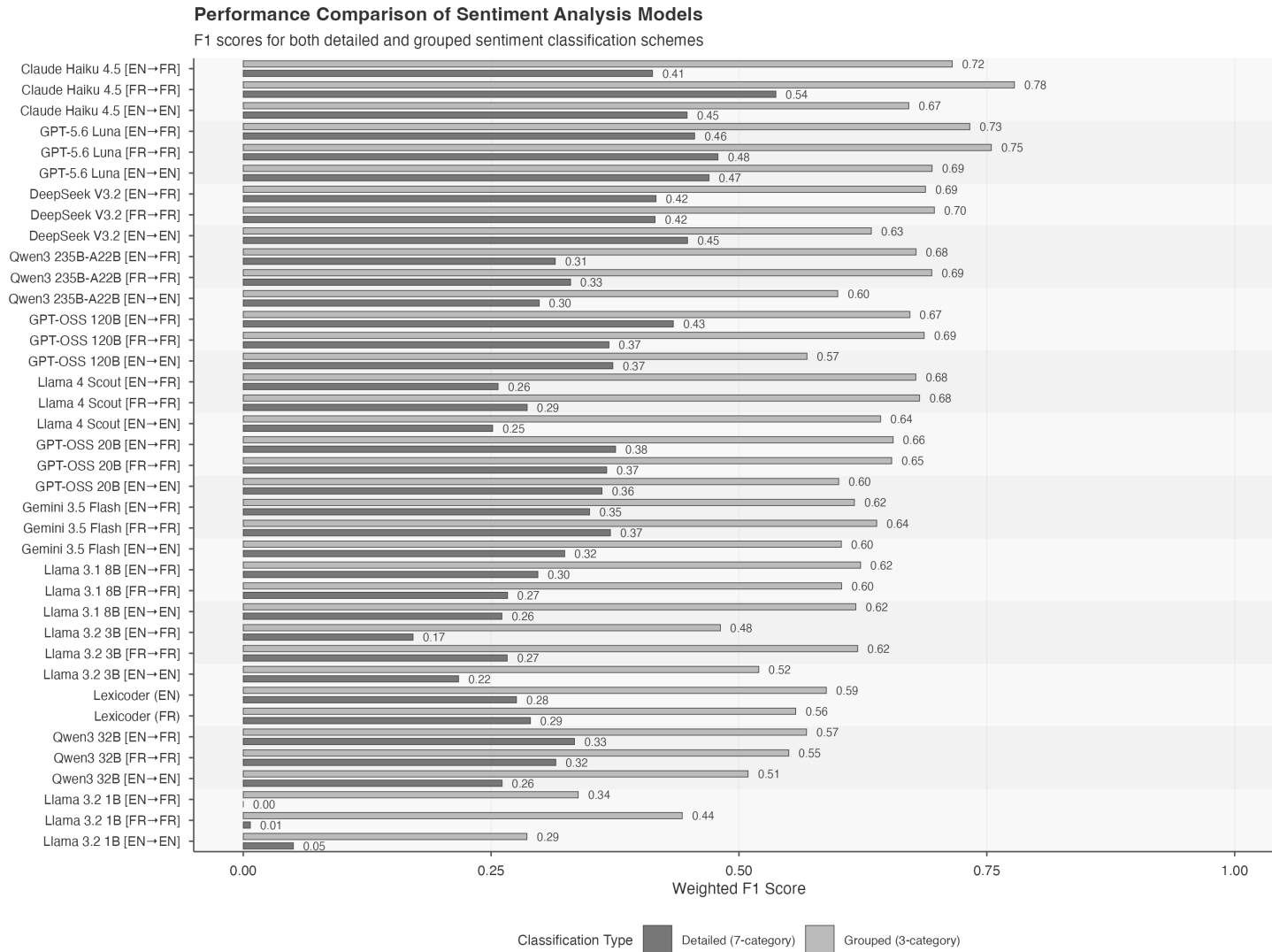


Figure 2. F1 scores for sentiment analysis across model architectures. Models grouped by F1 score performance and sorted by 3-category classification. Prompting mechanisms: EN→FR (English prompt on French text), FR→FR (French prompt on French text), and EN→EN (English prompt on translated text). Detailed classification includes 7 categories, grouped classification uses 3.

**Figure 3**  
*F1 Scores Comparison Across Models and Categories*

## Discussion

The central finding is that prompt language does not meaningfully affect LLM sentiment analysis performance on French text — a more nuanced picture than the English-prompting advantage reported elsewhere (Jiang et al., 2020; Lin et al., 2022; Muennighoff et al., 2023). If anything, the point estimates favored French prompts, and equivalence tests ruled out any English-prompt advantage larger than a few hundredths on every metric. What did cost performance was machine-translating the text itself: scoring English translations lowered correlation and 3-category F1 relative to working with the French original. This challenges the assumption that researchers must work in English to get good results on French text: francophone researchers can prompt in French with no analytical penalty, and teams can choose between French and English prompts by convenience rather than performance. The practical warning runs the other way: machine-translating a French corpus into English before analysis is not neutral. Theoretically, the case of French suggests that the “English prompt advantage” documented in prior work may be task-dependent, language-pair specific, or diminishing as models improve; whether it holds for other languages remains open.

This finding is only practically useful if LLMs are themselves viable for the task, and the results support that they are. Leading closed-weight models achieved strong correlations (GPT-5.6 Luna,  $r = 0.810$ ) and 3-category F1 scores (Claude Haiku 4.5, 0.778) without any task-specific fine-tuning, several configurations agreed with the ground truth more closely than individual human coders did, and the largest open-weight models came close to them (Qwen3 235B-A22B,  $r = 0.774$ ). Model size sets a floor, however. Models from roughly 8B parameters upward produced usable scores, a 3B model was unreliable across conditions, and a 1B model could not follow the instruction to return a number at all. Cost does not stand in the way: at list prices, scoring 1,000 sentences cost between USD 0.02 and 0.40 for every model but one (Table B1). The exception, Gemini 3.5 Flash (USD 3.77), reflects the reasoning tokens it generates by default rather than its per-token price. Together, these results mean social scientists can confidently deploy leading closed-weight LLMs, and larger open-weight ones, for sentiment analysis of French text with

French or English prompts, whichever suits their workflow, without fine-tuning expertise or heavy computational resources (Přibáň et al., 2024) — provided they avoid the smallest models.

Two secondary results qualify this picture. First, LLMs clearly outperformed the Lexicoder Sentiment Dictionary on correlation (mean LLM  $r = 0.673$  vs. 0.548 for the English dictionary) and, more modestly, on 3-category F1 (0.640 vs. 0.588), but the dictionaries' near-zero output gave them a slightly lower MAE — a reminder that MAE alone can reward a method that barely discriminates. Second, all models performed markedly worse on fine-grained 7-category classification than on 3-category polarity (mean F1 of 0.349 vs. 0.640), struggling most with extreme sentiment categories.

These two results are compatible once ordering and calibration are distinguished. LLMs capture sentiment intensity in the sense of ordering sentences from least to most positive and reproducing the spread of human ratings: their concordance with the ground truth (mean CCC = 0.631, up to 0.779) far exceeds that of the dictionaries (0.233 and 0.181), whose scores barely depart from zero. What LLMs do poorly is place scores on the right side of fixed category thresholds. They overstate extremes — 10.8% of LLM scores exceed  $\pm 0.66$ , against 1.5% of ground-truth values — so a sentence rated 0.5 by humans and 0.8 by a model is correctly ordered but misclassified. The ground truth contributes to the mismatch: as an average of three coders, it is pulled toward the center, leaving only three sentences in the two extreme categories. A transparent, fully interpretable dictionary therefore remains a defensible baseline for basic polarity detection but not for measuring intensity, and researchers needing fixed fine-grained categories should not expect general-purpose LLMs to deliver them out of the box, though Claude Haiku 4.5's stronger 7-category performance (0.537) suggests that calibration varies across models.

### Limitations

Several limitations qualify these findings. The 200-sentence benchmark is adequate for comparing models but thin at the extremes of the scale: only two sentences fall in the very negative category and one in the very positive, so class-level F1 scores for these categories are unstable. The benchmark is also demanding for humans: agreement between individual coders

was moderate (Krippendorff's  $\alpha = 0.597$ ), and averaging three coders, while it yields a more reliable ground truth, also pulls ratings toward the center and makes the extreme categories scarcer still. Finally, the findings concern French only. The corpus is French-language news on a single technical/political domain (open-source software), so neither the performance hierarchy nor the absence of a prompt-language effect can be assumed to extend to other languages, especially those less represented in LLM training data, or to other genres such as social media or political debate.

Methodologically, the EN→EN condition relies on Google Translate, and translation quality was deliberately not validated: the condition reproduces what a researcher who does not read English can do, namely translate automatically and analyze the result. Its lower performance should accordingly be read as the cost of that workflow rather than of translation as such. Model selection prioritized affordability and prominence over including every top-performing model globally, reflecting the perspective of researchers who need to analyze corpora of millions of sentences rather than a few hundred. A single standardized prompt was used across all models for comparability, which may understate peak performance achievable through model-specific prompt engineering; the need for a response-parsing function also indicates models did not always follow the requested output format exactly.

Costs are reported at list prices and count cached prompt tokens at the full input rate, so they are upper bounds; actual invoices depend on provider-specific caching discounts. More fundamentally, this evaluation is a snapshot: given how quickly LLM capabilities and provider offerings change — several models evaluated in an earlier version of this study were deprecated within eighteen months — the specific rankings reported here should be expected to age quickly, even as the higher-level finding regarding prompt language in French is likely to hold across model generations.

## Conclusion

For French text, prompt language does not constrain how social scientists work with LLMs. Francophone teams are free to prompt in French, and the choice between French and English prompts ceases to be a design decision that needs defending. Machine translation is

another matter: translating the corpus into English before analysis, a step that has often preceded computational work on non-English text, lowered performance here and loses its rationale.

That opens work this study could not do. Whether the absence of a prompt-language effect holds beyond French — particularly for languages less represented in training data, where the English-prompting advantage was first reported — is the natural next test, and this design ports to it directly: the same three conditions, the same manually annotated benchmark, a different target language. If it does, comparative projects spanning several languages could let each team prompt in its own language and pool the results.

Method choice remains a judgment about research goals rather than a ranking. Dictionary methods retain an advantage these results do not overturn — every score they produce can be traced to the words that generated it — and that transparency is worth more than accuracy in some research designs. What the evidence here settles is narrower and, for researchers working with French text, more useful: the language of the prompt is not among the things that need to be traded off.

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## **Appendix A**

Appendix B

Figure B1  
Sentiment Analysis Models Mean Absolute Error

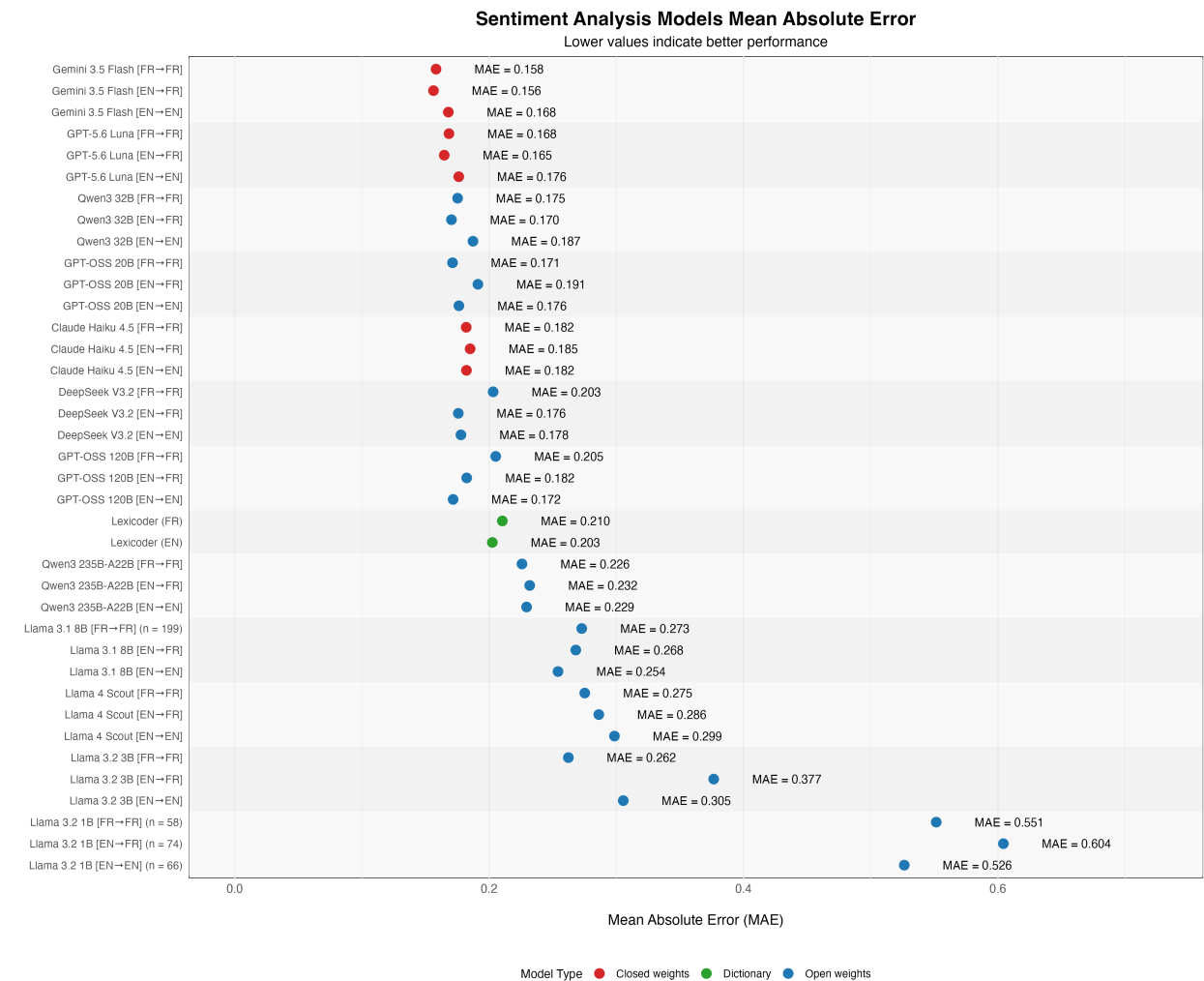


Figure 2. Cross-lingual sentiment analysis performance by Mean Absolute Error. Models grouped by type (color) and sorted by MAE (lower is better). Prompting mechanisms within clusters ordered as FR→FR (French prompt on French text), EN→FR (English prompt on French text), and EN→EN (English prompt on translated text).

Figure 4 shows the Mean Absolute Error (MAE) results across models and conditions, providing complementary insights into prediction accuracy beyond correlation analysis.

**Figure B2**  
*Sentiment Analysis Comparison Over Time*

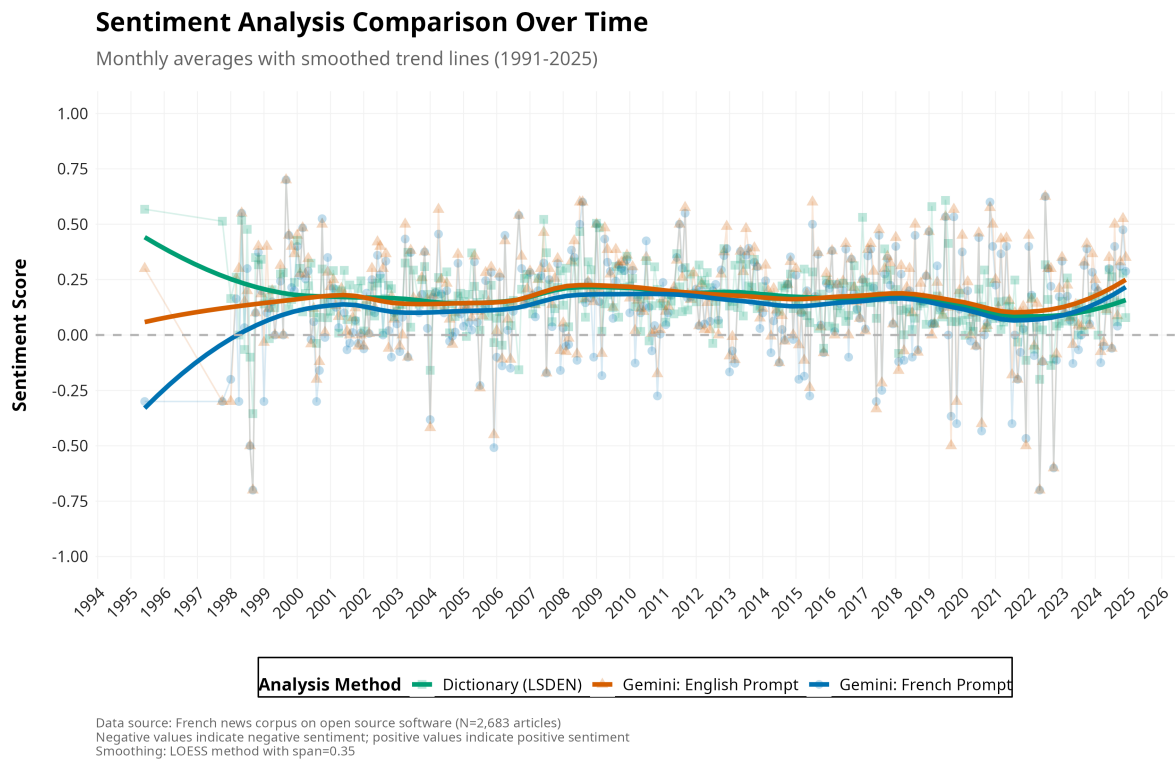


Figure 5 compares corpus-wide temporal sentiment trends captured by the best-performing LLM and by the Lexicoder Sentiment Dictionary, illustrating that the two methods track closely despite the LLM's higher correlation with sentence-level manual annotations.

**Table B1**

*Cost of Scoring 1,000 Sentences at List Prices (Upper Bound; Cached Prompt Tokens Counted at the Full Input Rate)*

| Model            | Provider  | Weights | API calls per usable score | Input tokens | Output tokens | USD per 1,000 sentences |
|------------------|-----------|---------|----------------------------|--------------|---------------|-------------------------|
| GPT-20B          | OpenRoute | Open    | 1.02                       | 657,496      | 87,892        | 0.018                   |
| Llama 3.2 3B     | OpenRoute | Open    | 1.00                       | 595,142      | 7,320         | 0.018                   |
| GPT-120B         | OpenRoute | Open    | 1.00                       | 667,425      | 95,660        | 0.020                   |
| Llama 3.2 1B     | OpenRoute | Open    | 22.68                      | 232,695      | 168,789       | 0.022                   |
| Qwen3 32B        | OpenRoute | Open    | 1.00                       | 592,086      | 5,696         | 0.027                   |
| Qwen3 235B-A22B  | OpenRoute | Open    | 1.00                       | 584,886      | 7,752         | 0.030                   |
| Llama 4 Scout    | OpenRoute | Open    | 1.00                       | 564,947      | 8,467         | 0.059                   |
| DeepSeek V3.2    | OpenRoute | Open    | 1.00                       | 559,594      | 9,340         | 0.067                   |
| Llama 3.1 8B     | OpenRoute | Open    | 1.01                       | 593,398      | 8,062         | 0.074                   |
| GPT-5.6 Luna     | OpenAI    | Closed  | 1.00                       | 552,225      | 51,819        | 0.096                   |
| Claude Haiku 4.5 | Anthropic | Closed  | 1.00                       | 648,705      | 12,600        | 0.395                   |
| Gemini 3.5 Flash | Google    | Closed  | 1.03                       | 568,160      | 658,571       | 3.766                   |

**Table B2***F1 Scores for 3-Category Classification*

| Model                    | negative | neutral | positive | weighted_f1 |
|--------------------------|----------|---------|----------|-------------|
| Claude Haiku 4.5 (FR→FR) | 0.762    | 0.375   | 0.832    | 0.778       |
| GPT-5.6 Luna (FR→FR)     | 0.748    | 0.385   | 0.8      | 0.755       |
| GPT-5.6 Luna (EN→FR)     | 0.73     | 0.364   | 0.776    | 0.733       |
| Claude Haiku 4.5 (EN→FR) | 0.69     | 0.273   | 0.778    | 0.715       |
| DeepSeek V3.2 (FR→FR)    | 0.672    | 0.216   | 0.765    | 0.697       |
| GPT-5.6 Luna (EN→EN)     | 0.688    | 0.367   | 0.736    | 0.695       |
| Qwen3 235B-A22B (FR→FR)  | 0.752    | 0.243   | 0.716    | 0.695       |
| DeepSeek V3.2 (EN→FR)    | 0.696    | 0.226   | 0.737    | 0.688       |
| GPT-OSS 120B (FR→FR)     | 0.721    | 0.314   | 0.712    | 0.687       |
| Llama 4 Scout (FR→FR)    | 0.643    | 0.291   | 0.747    | 0.682       |
| Qwen3 235B-A22B (EN→FR)  | 0.719    | 0.254   | 0.706    | 0.679       |
| Llama 4 Scout (EN→FR)    | 0.667    | 0.258   | 0.732    | 0.678       |
| GPT-OSS 120B (EN→FR)     | 0.717    | 0.324   | 0.689    | 0.672       |
| Claude Haiku 4.5 (EN→EN) | 0.678    | 0.256   | 0.715    | 0.671       |
| GPT-OSS 20B (EN→FR)      | 0.7      | 0.268   | 0.677    | 0.655       |
| GPT-OSS 20B (FR→FR)      | 0.691    | 0.258   | 0.68     | 0.654       |
| Llama 4 Scout (EN→EN)    | 0.647    | 0.246   | 0.686    | 0.643       |
| Gemini 3.5 Flash (FR→FR) | 0.655    | 0.264   | 0.673    | 0.639       |
| DeepSeek V3.2 (EN→EN)    | 0.672    | 0.258   | 0.657    | 0.633       |
| Llama 3.1 8B (EN→FR)     | 0.652    | 0.281   | 0.646    | 0.623       |
| Llama 3.2 3B (FR→FR)     | 0.465    | 0.195   | 0.747    | 0.62        |
| Llama 3.1 8B (EN→EN)     | 0.627    | 0.265   | 0.653    | 0.618       |
| Gemini 3.5 Flash (EN→FR) | 0.642    | 0.242   | 0.646    | 0.616       |
| Llama 3.1 8B (FR→FR)     | 0.615    | 0.174   | 0.643    | 0.604       |
| Gemini 3.5 Flash (EN→EN) | 0.667    | 0.233   | 0.613    | 0.603       |
| GPT-OSS 20B (EN→EN)      | 0.614    | 0.234   | 0.635    | 0.601       |
| Qwen3 235B-A22B (EN→EN)  | 0.698    | 0.227   | 0.591    | 0.6         |
| Lexicoder (EN)           | 0.574    | 0.231   | 0.636    | 0.588       |
| GPT-OSS 120B (EN→EN)     | 0.678    | 0.218   | 0.552    | 0.569       |
| Qwen3 32B (EN→FR)        | 0.554    | 0.165   | 0.621    | 0.568       |
| Lexicoder (FR)           | 0.474    | 0.289   | 0.631    | 0.557       |
| Qwen3 32B (FR→FR)        | 0.515    | 0.177   | 0.611    | 0.55        |
| Llama 3.2 3B (EN→EN)     | 0.167    | 0.261   | 0.73     | 0.52        |
| Qwen3 32B (EN→EN)        | 0.559    | 0.19    | 0.52     | 0.509       |
| Llama 3.2 3B (EN→FR)     | 0        | 0.222   | 0.757    | 0.481       |
| Llama 3.2 1B (FR→FR)     | 0        | 0.138   | 0.648    | 0.443       |
| Llama 3.2 1B (EN→FR)     | 0        | 0       | 0.568    | 0.338       |
| Llama 3.2 1B (EN→EN)     | 0        | 0.15    | 0.457    | 0.286       |

**Table B3***F1 Scores for 7-Category Classification*

| <b>Model</b>             | <b>very_neg</b> | <b>negative</b> | <b>somewhat_neg</b> | <b>neutral</b> | <b>somewhat_pos</b> | <b>positive</b> | <b>very_pos</b> | <b>weighted</b> |
|--------------------------|-----------------|-----------------|---------------------|----------------|---------------------|-----------------|-----------------|-----------------|
| Claude Haiku 4.5 (FR→FR) | 0.4             | 0.68            | 0.455               | 0.375          | 0.54                | 0.612           | 0               | 0.537           |
| GPT-5.6 Luna (FR→FR)     | 0.286           | 0.5             | 0.393               | 0.385          | 0.538               | 0.462           | 0.133           | 0.479           |
| GPT-5.6 Luna (EN→EN)     | 0.364           | 0.5             | 0.338               | 0.367          | 0.51                | 0.542           | 0.25            | 0.47            |
| GPT-5.6 Luna (EN→FR)     | 0.333           | 0.5             | 0.258               | 0.364          | 0.564               | 0.407           | 0.125           | 0.455           |
| DeepSeek V3.2 (EN→EN)    | 0.333           | 0.533           | 0.419               | 0.258          | 0.415               | 0.588           | 0.667           | 0.448           |
| Claude Haiku 4.5 (EN→EN) | 0.286           | 0.642           | 0.4                 | 0.256          | 0.409               | 0.568           | 0               | 0.448           |
| GPT-OSS 120B (EN→FR)     | 0.333           | 0.566           | 0.4                 | 0.324          | 0.416               | 0.485           | 0.133           | 0.434           |
| DeepSeek V3.2 (EN→FR)    | 0.444           | 0.667           | 0.429               | 0.226          | 0.349               | 0.493           | 0               | 0.416           |
| DeepSeek V3.2 (FR→FR)    | 0.333           | 0.368           | 0.469               | 0.216          | 0.4                 | 0.527           | 0.167           | 0.415           |
| Claude Haiku 4.5 (EN→FR) | 0.5             | 0.68            | 0.364               | 0.273          | 0.347               | 0.511           | 0.2             | 0.413           |
| GPT-OSS 20B (EN→FR)      | 0.235           | 0.431           | 0.269               | 0.268          | 0.37                | 0.535           | 0               | 0.376           |
| GPT-OSS 120B (EN→EN)     | 0.444           | 0.692           | 0.281               | 0.218          | 0.288               | 0.536           | 0.286           | 0.373           |
| Gemini 3.5 Flash (FR→FR) | 0.222           | 0.583           | 0.25                | 0.264          | 0.31                | 0.571           | 0.2             | 0.37            |
| GPT-OSS 120B (FR→FR)     | 0.308           | 0.545           | 0.333               | 0.314          | 0.319               | 0.447           | 0.154           | 0.369           |
| GPT-OSS 20B (FR→FR)      | 0.364           | 0.6             | 0.286               | 0.258          | 0.325               | 0.448           | 0.286           | 0.367           |
| GPT-OSS 20B (EN→EN)      | 0.4             | 0.528           | 0.235               | 0.234          | 0.338               | 0.508           | 0               | 0.362           |
| Gemini 3.5 Flash (EN→FR) | 0.4             | 0.694           | 0.255               | 0.242          | 0.263               | 0.485           | 0.167           | 0.349           |
| Qwen3 32B (EN→FR)        | 0.222           | 0.512           | 0.163               | 0.165          | 0.286               | 0.615           | 0               | 0.334           |
| Qwen3 235B-A22B (FR→FR)  | 0.19            | 0.407           | 0.32                | 0.243          | 0.319               | 0.369           | 0.087           | 0.33            |
| Gemini 3.5 Flash (EN→EN) | 0.364           | 0.652           | 0.222               | 0.233          | 0.23                | 0.5             | 0               | 0.324           |
| Qwen3 32B (FR→FR)        | 0.333           | 0.558           | 0.125               | 0.177          | 0.236               | 0.632           | 0               | 0.315           |
| Qwen3 235B-A22B (EN→FR)  | 0.182           | 0.321           | 0.24                | 0.254          | 0.35                | 0.339           | 0.091           | 0.315           |
| Qwen3 235B-A22B (EN→EN)  | 0.182           | 0.275           | 0.302               | 0.227          | 0.279               | 0.408           | 0.133           | 0.299           |
| Llama 3.1 8B (EN→FR)     | 0.111           | 0.12            | 0.192               | 0.281          | 0.319               | 0.507           | 0.2             | 0.297           |
| Lexicoder (FR)           | 0               | 0               | 0.253               | 0.289          | 0.479               | 0.057           | 0               | 0.29            |
| Llama 4 Scout (FR→FR)    | 0.125           | 0.167           | 0.182               | 0.291          | 0.341               | 0.356           | 0.1             | 0.286           |
| Lexicoder (EN)           | 0               | 0               | 0.267               | 0.231          | 0.472               | 0               | 0               | 0.276           |
| Llama 3.1 8B (FR→FR)     | 0.143           | 0.197           | 0.179               | 0.174          | 0.239               | 0.543           | 0               | 0.267           |
| Llama 3.2 3B (FR→FR)     | 0               | 0.364           | 0.2                 | 0.195          | 0.24                | 0.395           | 0               | 0.266           |
| Qwen3 32B (EN→EN)        | 0.222           | 0.364           | 0.172               | 0.19           | 0.168               | 0.567           | 0.333           | 0.261           |
| Llama 3.1 8B (EN→EN)     | 0.138           | 0.222           | 0.136               | 0.265          | 0.246               | 0.479           | 0.143           | 0.261           |
| Llama 4 Scout (EN→FR)    | 0.154           | 0.261           | 0.143               | 0.258          | 0.281               | 0.329           | 0.08            | 0.257           |
| Llama 4 Scout (EN→EN)    | 0.105           | 0.28            | 0.125               | 0.246          | 0.234               | 0.434           | 0.154           | 0.252           |